

Muhammad Ahmed Rayyan

+92 315 2097769 | ahmedrayyanfamily@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI and Machine Learning student passionate about building intelligent systems that solve real-world problems. My work spans healthcare AI at AKUH, where I developed pediatric MRI segmentation models, to winning ML competitions and deploying LLM-based applications. I thrive at the intersection of research and implementation, turning complex data into functional, impactful solutions.

EDUCATION

Bachelor of Science Artificial Intelligence

Shaheed Zulfikar Ali Bhutto Institute of Science and Technology (SZABIST), Karachi

Sept 2023 - Present

GPA: 3.32 / 4.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Artificial Neural Networks, Data Structures & Algorithms, Design & Analysis of Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Agent-Based Modeling

Intermediate (Pre-Engineering)

Govt. Degree Science & Commerce College, Gulshan-e-Iqbal Block-7, Karachi

Jul 2021 - Jul 2023

TECHNICAL SKILLS

Programming & Web: Python, C/C++, Java, JavaScript, HTML5/CSS3, Flask, Streamlit, Gradio, NodeJS

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, NLTK, nnU-Net, OpenCV, Stata

Tools & Platforms: Git/GitHub, n8n, MySQL, Microsoft SQL Server, Arduino IDE, Raspberry Pi, NetLogo, Netlify

WORK EXPERIENCE

Aga Khan University Hospital (AKUH)

AI/ML Intern | Python, PyTorch, nnU-Net, NumPy, Pandas

Karachi, Pakistan

Jul 2025 – Aug 2025

- Implemented nnU-Net for pediatric MRI segmentation (LISA Challenge), achieving mean Dice of 0.68 and IoU of 0.52.
- Conducted research on MRI super-resolution and deep learning segmentation techniques for medical imaging applications.
- Evaluated model outputs through quality control and explored emerging architectures such as MAMBA-based segmentation models.

PROJECTS

VitalCare-CustomGPT | Python, Streamlit, Hugging Face, Transformers | [Github](#)

- Built a multi-role medical AI assistant with fine-tuned Hugging Face models for domain-specific healthcare responses.
- Implemented role-based prompt engineering and keyword validation to enhance response relevance and medical domain accuracy.
- Developed an interactive chat interface with Streamlit, enabling real-time LLM inference and user-friendly medical assistance.

EEG Brain Signal Classifier | Python, scikit-learn, NumPy, Joblib | [Github](#)

- Developed a machine learning pipeline to classify EEG brain signals into actionable commands using feature extraction techniques.
- Implemented KNN and Random Forest classifiers with cross-validation for model selection and performance optimization.
- Processed EEG datasets stored in JSON format, transforming multi-channel temporal signals into ML-ready feature representations.

MediQuery-RAG | Python, Streamlit, LangChain, ChromaDB | [Github](#)

- Built a RAG pipeline (984 chunks, 8 FDA labels) with Groq's Llama 3.1 8B, delivering citation-backed, zero-hallucination responses.
- Evaluated via LLM-as-judge across 20 queries, achieving Faithfulness 0.82 and Answer Relevancy 0.78 (RAGAS-equivalent).
- Engineered a rephrase-then-retrieve chain with conversational memory to resolve multi-turn follow-up queries accurately.

ACHIEVEMENTS & ACTIVITIES

First Prize – Machine Learning Competition

ZABEFEST, SZABIST | 2025

Developed an XGBoost-based pipeline to predict liver cirrhosis patient survival outcomes (Mayo Clinic dataset, 17 clinical features) using multi-class classification. Achieved a log-loss of 0.3557, securing 1st place on the competition leaderboard.

3rd Prize – AI Image Generation Competition (World No Tobacco Day)

Community Health Sciences Department, Aga Khan University | 2025

Created AI-generated public health awareness imagery discouraging tobacco usage, demonstrating creative application of AI for social impact and awareness. [Virtual Gallery](#)

Member – Tobacco Free Cities Coalition Initiative

Community Health Sciences Department, Aga Khan University | 2025

Contributed to public health awareness initiatives promoting tobacco-free environments through collaborative activities.